Pipelined ECC Hardware Results at a Common Operating Point

W = 64 · 250 MHz · Target bandwidth: 3,072 GB/s · UMC 28 nm pre-layout cell area

Configuration	Block	Single-engine area	Single-engine throughput	Engines required for 3.072 TB/s	Total replicated area	Total NAND2 gate equivalents
BCH(k=4096, t=12)	Encoder	2,754.0 µm²	1.910 GB/s	1,608	4.428 mm²	8.79 M
	Decoder	39,834.6 µm²	1.855 GB/s	1,656	65.966 mm²	130.89 M
PBCH(k=4096, t=8, l=3)	Encoder	7,385.3 µm²	1.792* GB/s	1,715	12.666 mm²	25.13 M
	Decoder	27,545.3 µm²	1.855 GB/s	1,656	45.615 mm²	90.51 M
BCH(k=4096, t=28)	Encoder	5,656.4 µm²	1.829 GB/s	1,680	9.503 mm²	18.85 M
	Decoder	121,274.3 µm²	1.778 GB/s	1,728	209.562 mm²	415.80 M
PBCH(k=4096, t=8, l=15)	Encoder	11,163.8 µm²	1.419* GB/s	2,166	24.181 mm²	47.98 M
	Decoder	29,250.1 µm²	1.855 GB/s	1,656	48.438 mm²	96.11 M

*: PBCH 인코더 throughput은 BER_defect로 인한 평균 defect 수(l=3: 0.4203개, l=15: 4.215개/codeword) 기준

Throughput: steady-state codeword 입력 간격 기준

Area: UMC 28 nm pre-layout cell area · NAND2 GE: STP_ND2_0P5 = 0.504 µm²